Designing a 2-Bit Up/Down Counter

Behavioral, Gate-Level, and Structural Modeling in Verilog

Overview

In this lab you will design the same digital circuit — a synchronous 2-bit up/down counter — three separate times, once in each of the three major Verilog modeling styles: behavioral, gate-level, and structural. The goal is not just to get a working circuit, but to understand how the same specification maps onto very different descriptions, and what each style costs you or buys you in terms of readability, control, and reuse.

Learning Objectives

- Translate a functional specification and truth table into three different Verilog implementations of the same circuit.
- Derive minimal Boolean next-state equations by hand and implement them using primitive logic gates.
- Build a structural (hierarchical) design out of instantiated sub-modules, including a reusable D flip-flop module.
- Use a Verilog function to compute a value and a task to organize repeated testbench actions.
- Compare the three styles and reason about trade-offs between them.

Deliverables

1. Three Verilog source files: counter_behavioral.v, counter_gatelevel.v, counter_structural.v.
2. One shared testbench, counter_tb.v, that can test any of the three.
3. Simulation waveform screenshot and console log showing correct counting in both directions and correct reset behavior.

Design Specification

You will design a module with the following interface. Use exactly this name and port list in all three parts so that the shared testbench works without modification.

```
module counter2 (
    input  clock,
    input  reset,   // asynchronous, active-high
    input  up,      // 1 = count up, 0 = count down
    output [1:0] count
);
```